

Moritz-Alexander Thiel

PHYSICS GRADUATE

ABOUT ME

I am a curious and analytic physics graduate, that loves to solve complex problems. Seeking to apply advanced physics knowledge in a dynamic research or industry role, aiming to drive innovation and contribute to cutting-edge projects.

EXPERIENCE

RESEARCHER

Since February 2025 | Bauhaus Luftfahrt e.V.

- Simulations driven by scenarios on the future of aviation up to 2070 with a focus on fuel and hydrogen
- Modeling of the fuel production process and chemical procedures within the Sun-to-Liquid-II project, as well as analysis of its system integration potentials

GRADUATE STUDENT RESEARCHER

May 2023 – July 2024 | Technical University Munich (TUM)

- Conducted fundamental research to improve fuel cells by using electrochemical erosion for platinum nanoparticle synthesis in a clean and environmental friendly way
- Execution and Evaluation of up-scaling techniques for bringing this procedure from laboratory scales towards serial applications
- Achieved 17x output of synthesized highly active catalyst material while keeping long-term stability

STUDENT RESEARCH ASSISTANT

Sep 2022 – Feb 2023 | Technical University Munich (TUM)

- Leading a project to build an autonomous solution for analyzing magnetic fields for particle detection
- Responsible for the conception, planning and technical implementation of the sensor API via python

INTERN

Sep 2019 – Sep 2019 | solid GmbH

- Implemented the front- and backend of a new internal incident management tool (via VBA)
- Achieved to automate the handling of incidents, which drastically decreased errors as well as accelerated the process due to, e.g. automatic due date mails

MYSTERY SHOPPER



CONTACT

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[LINKEDIN](#)

SKILLS

Microsoft Office and **LaTeX**
Programming (Python, SQL, VBA)
Data Analysis (Origin, Numpy, Pandas)
CAD (Onshape)
Git

LANGUAGES

German (Native)
English (C2)
Latin (Latinum)
Ancient Greek (Graecum)

HOBBIES

3D Printing
Vibe Coding (Cursor, Claude Code)
Politics
Personal Development (Reading, News, Podcasts)
Culture (Theater, Art, Travelling)

Oct 2017 – Jan 2019 | aha! TALENTEXPERTS

- Tested and evaluated the success of employee training programs in the travel market
 - Further shaped these training programs by providing qualitative and quantitative feedback as well as improvement suggestions
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EDUCATION

APPLIED AND ENGINEERING PHYSICS (M.SC.)

Apr 2021 – July 2024 | Technical University Munich (TUM)

- **Grade:** 1.6
- **Master Thesis:** Electrochemical Erosion of Supported Platinum Nanoparticles for the Oxygen Reduction Reaction: Optimization and Upscaling
- **Focus:** Renewable energies, data science

PHYSICS (M.SC.)

Sep 2021 – Dec 2021 | University of California, San Diego

- Exchange Semester in the US
- **Focus:** Modern statistics, aerospace technologies

PHYSICS (B.SC.)

Oct 2017 – Sep 2021 | Technical University Munich (TUM)

- **Bachelor Thesis:** Testing the Fermi/GBM response matrix at its limits
- **Focus:** Physics foundations, advanced mathematics, quantum mechanics, thermodynamics
- Scholarship at the "Friedrich-Naumann-Stiftung für die Freiheit"

ABITUR

Sep 2009 – Aug 2017 | Melanchthon-Gymnasium Nürnberg

- **Grade:** 1.4
- Part of the program for highly skilled pupils (Hochbegabten-Modell-Mittelfranken) from 2013 until 2016